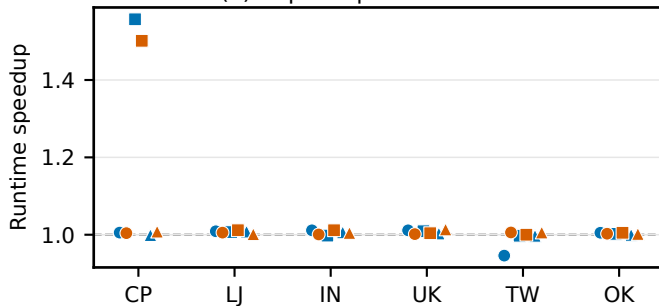
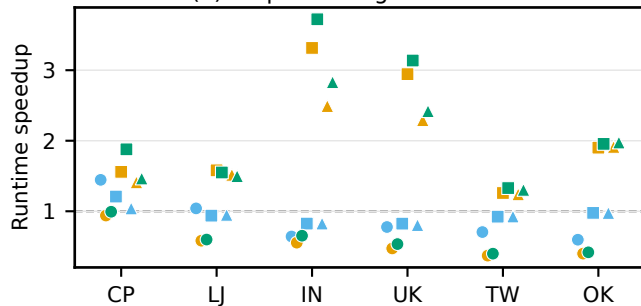


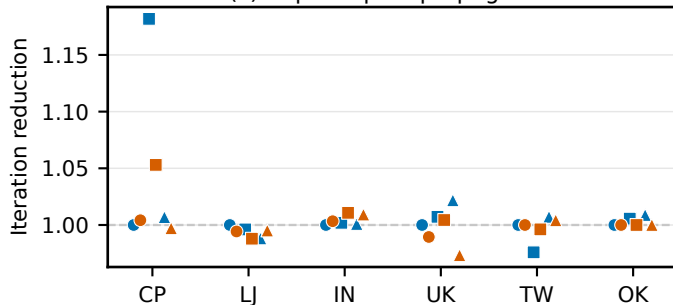
(a) In-place pull: end-to-end



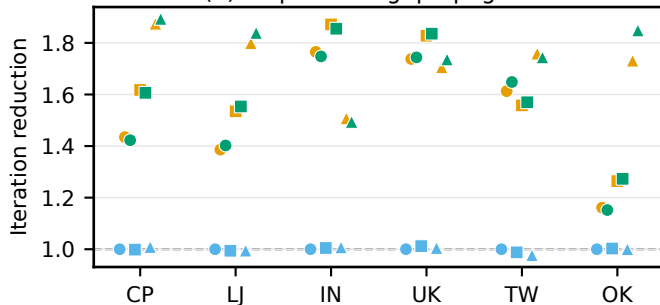
(b) Preprocessing: end-to-end



(c) In-place pull: propagation



(d) Preprocessing: propagation



● Original graph

● Reorder only

● Full preprocessing

■ SSSP

● Preprocessed graph

● Shortcut only

● BFS

▲ SSWP